AMRITA VISHWA VIDYAPEETHAM

Full Stack Development

Lab 2-2-2026

Name:Aditya K U
Roll:AV.SC.U4AIE23001
AIE-A

Bootstrap

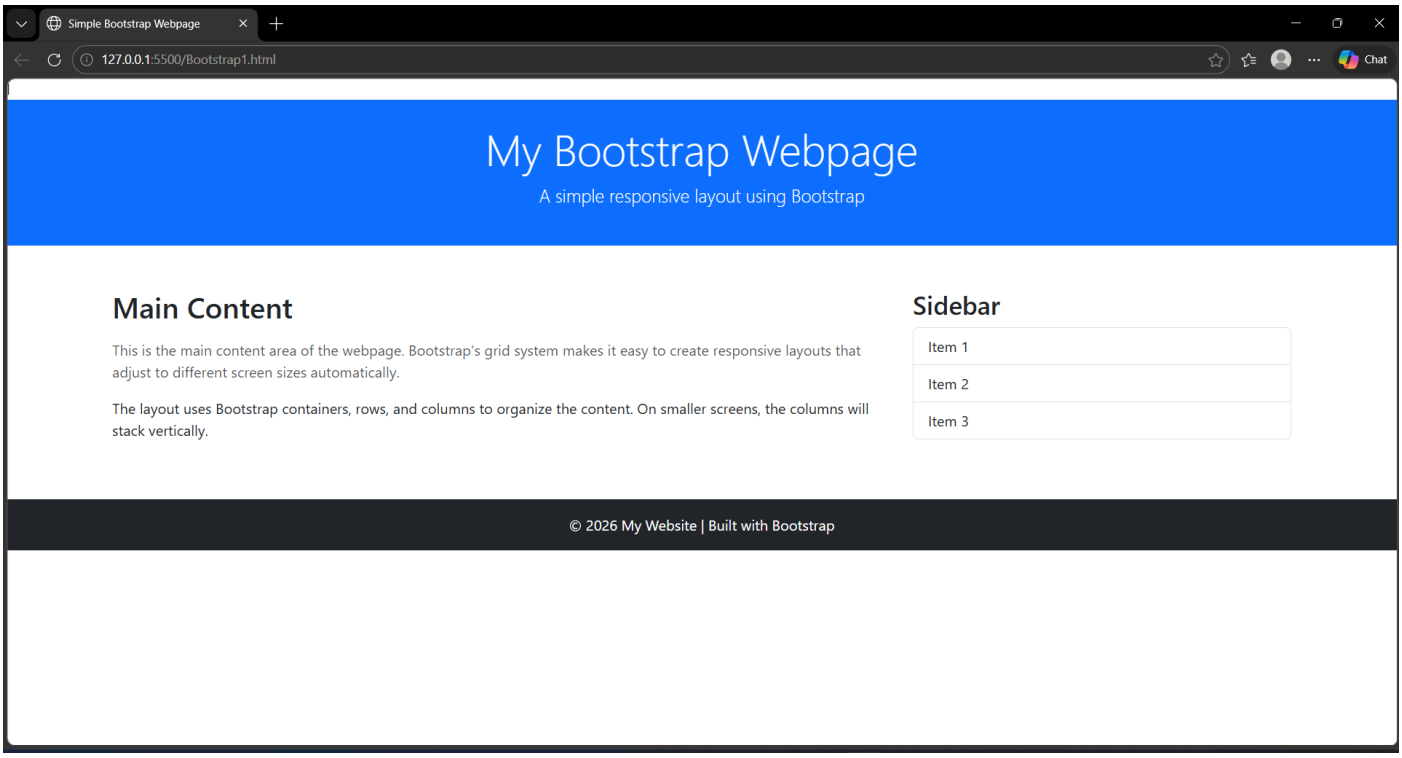
1. Write a program to create a simple responsive webpage with a header, main content area, and a footer using Bootstrap's grid system, containers, and typography classes.

HTML Code

```
l<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Simple Bootstrap Webpage</title>

  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>

<body>

  <header class="bg-primary text-white text-center py-4">
    <div class="container">
      <h1 class="display-5">My Bootstrap Webpage</h1>
      <p class="lead">A simple responsive layout using Bootstrap</p>
    </div>
  </header>

  <main class="container my-5">

    <div class="row">

      <div class="col-md-8">
        <h2 class="mb-3">Main Content</h2>

        <p class="text-muted">
          This is the main content area of the webpage. Bootstrap's grid system
          makes it easy to create responsive layouts that adjust to different
          screen sizes automatically.
        </p>

        <p>
          The layout uses Bootstrap containers, rows, and columns to organize
          the content. On smaller screens, the columns will stack vertically.
        </p>
```

```
        </div>

        <div class="col-md-4">
          <h3>Sidebar</h3>

          <ul class="list-group">
            <li class="list-group-item">Item 1</li>
            <li class="list-group-item">Item 2</li>
            <li class="list-group-item">Item 3</li>
          </ul>
        </div>

      </div>

    </main>

    <footer class="bg-dark text-white text-center py-3">
      <div class="container">
        <p class="mb-0">© 2026 My Website | Built with Bootstrap</p>
      </div>
    </footer>

    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>

  </body>
</html>
```

Output:



2. Write a program to build a responsive product gallery using Bootstrap's cards and buttons, where each card displays an image, title, description, and a call-to-action button.

HTML Code

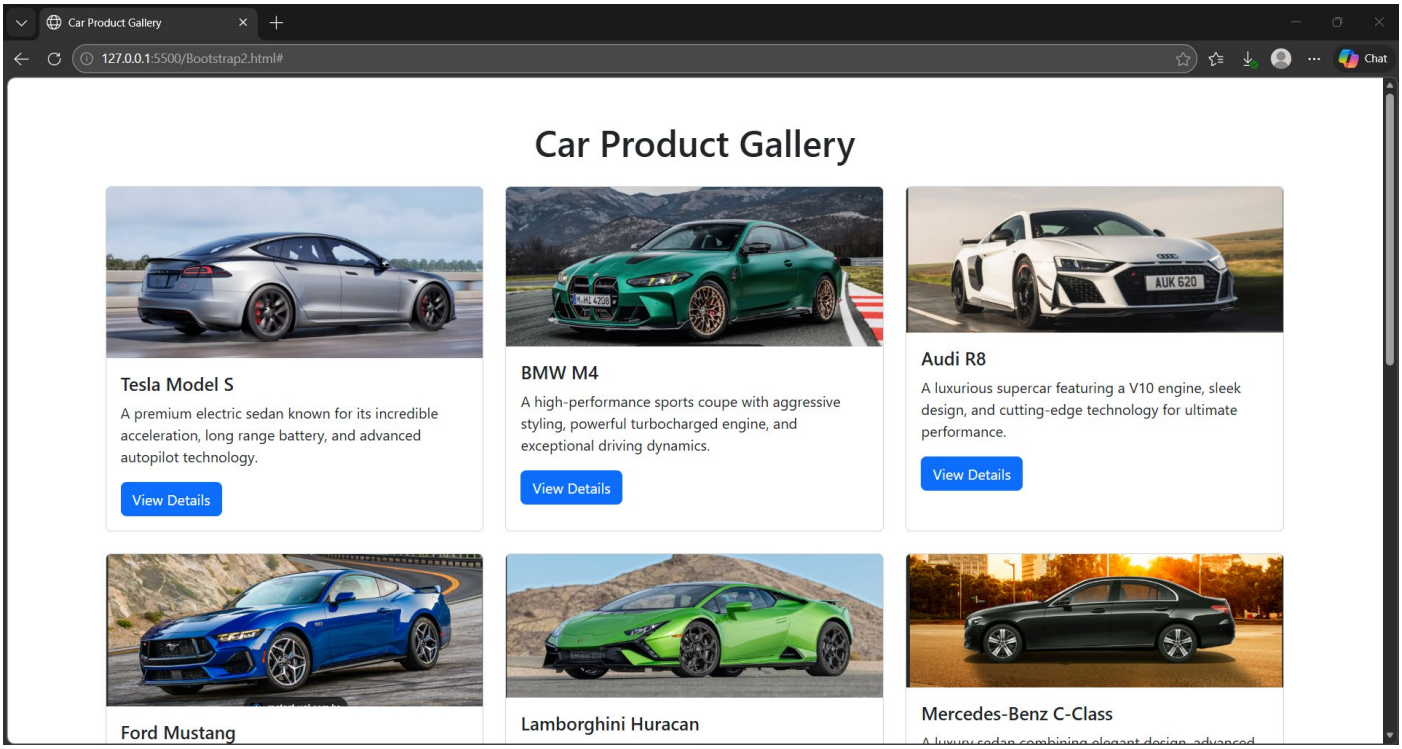
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Car Product Gallery</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

</head>

<body>

<div class="container my-5">

<h1 class="text-center mb-4">Car Product Gallery</h1>

<div class="row g-4">

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="1.jpg" class="card-img-top" alt="Tesla Model S">
<div class="card-body">
<h5 class="card-title">Tesla Model S</h5>
<p class="card-text">
A premium electric sedan known for its incredible acceleration, long range battery, and advanced autopilot technology.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="2.jpg" class="card-img-top" alt="BMW M4">
<div class="card-body">
<h5 class="card-title">BMW M4</h5>
<p class="card-text">
```

```
A high-performance sports coupe with aggressive styling, powerful turbocharged engine, and exceptional driving dynamics.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="3.jpg" class="card-img-top" alt="Audi R8">
<div class="card-body">
<h5 class="card-title">Audi R8</h5>
<p class="card-text">
A luxurious supercar featuring a V10 engine, sleek design, and cutting-edge technology for ultimate performance.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="4.jpg" class="card-img-top" alt="Ford Mustang">
<div class="card-body">
<h5 class="card-title">Ford Mustang</h5>
<p class="card-text">
An iconic American muscle car known for its bold design, powerful engine options, and thrilling driving experience.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="5.jpg" class="card-img-top" alt="Lamborghini Huracan">
<div class="card-body">
<h5 class="card-title">Lamborghini Huracan</h5>
<p class="card-text">
A stunning Italian supercar offering extreme speed, aerodynamic design, and a roaring V10 engine.
</p>
<a href="#" class="btn btn-primary">View Details</a>
```

```
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="6.jpg" class="card-img-top" alt="Mercedes Benz C-Class">
<div class="card-body">
<h5 class="card-title">Mercedes-Benz C-Class</h5>
<p class="card-text">
A luxury sedan combining elegant design, advanced safety features, and a comfortable driving experience.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="7.jpg" class="card-img-top" alt="Toyota Supra">
<div class="card-body">
<h5 class="card-title">Toyota Supra</h5>
<p class="card-text">
A legendary sports car featuring a turbocharged engine, aerodynamic design, and outstanding handling.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="8.jpg" class="card-img-top" alt="Porsche 911">
<div class="card-body">
<h5 class="card-title">Porsche 911</h5>
<p class="card-text">
A timeless sports car with precise engineering, powerful performance, and unmatched driving pleasure.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
```

```
<img src="9.jpg" class="card-img-top" alt="Ferrari F8">
<div class="card-body">
<h5 class="card-title">Ferrari F8</h5>
<p class="card-text">
A high-performance Italian supercar delivering breathtaking speed, luxury, and advanced aerodynamics.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

<div class="col-md-4 col-sm-6">
<div class="card h-100">
<img src="10.jpg" class="card-img-top" alt="Range Rover Sport">
<div class="card-body">
<h5 class="card-title">Range Rover Sport</h5>
<p class="card-text">
A premium SUV known for its off-road capability, luxurious interior, and powerful engine options.
</p>
<a href="#" class="btn btn-primary">View Details</a>
</div>
</div>
</div>

</div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```

Output:



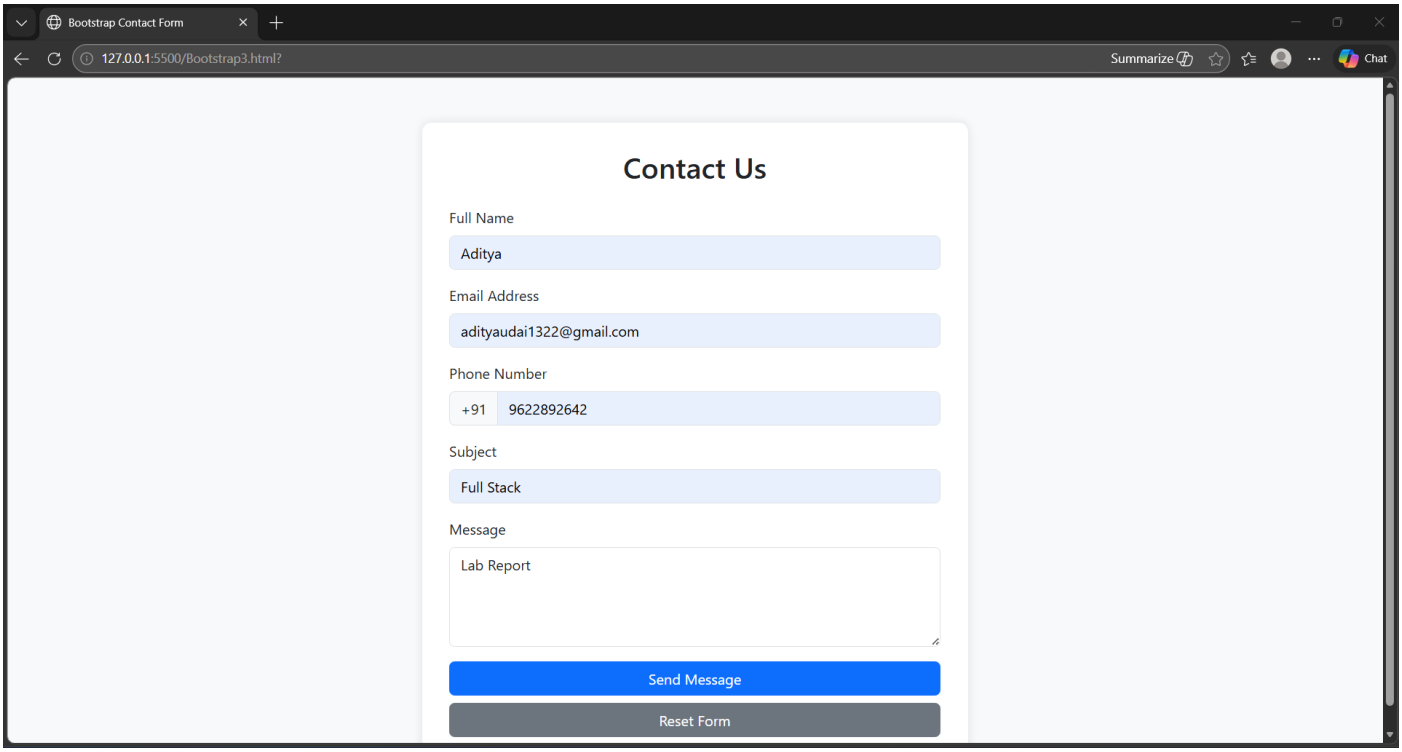
3. Write a program to design a contact form with different input types (text, email, textarea) and use Bootstrap's form-control and input-group classes to style and organize the form fields.

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Bootstrap Contact Form</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

</head>
<body>
<div class="container">
<div class="form-container">
<h2 class="text-center mb-4">Contact Us</h2>
<form>
<div class="mb-3">
<label class="form-label">Full Name</label>
<input type="text" class="form-control" placeholder="Enter your name">
</div>

<div class="mb-3">
<label class="form-label">Email Address</label>
<input type="email" class="form-control" placeholder="Enter your email">
```

```
</div>

<div class="mb-3">
<label class="form-label">Phone Number</label>

<div class="input-group">
<span class="input-group-text">+91</span>
<input type="text" class="form-control" placeholder="Enter phone number">
</div>

</div>
<div class="mb-3">
<label class="form-label">Subject</label>
<input type="text" class="form-control" placeholder="Enter subject">
</div>
<div class="mb-3">
<label class="form-label">Message</label>
<textarea class="form-control" rows="4" placeholder="Write your message here"></textarea>
</div>
<div class="d-grid gap-2">

<button type="submit" class="btn btn-primary">
Send Message
</button>

<button type="reset" class="btn btn-secondary">
Reset Form
</button>
</div>
</form>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Output:



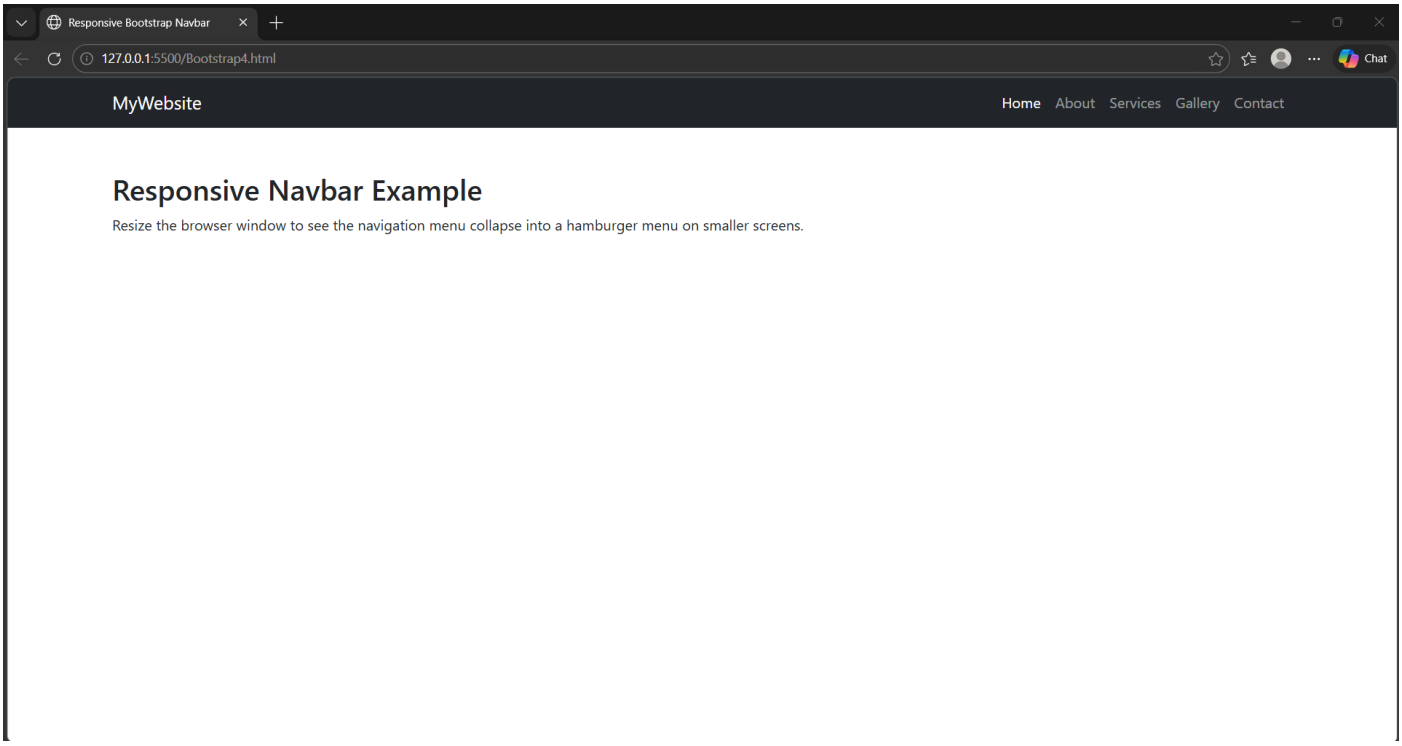
4. Write a program to implement a responsive navigation bar in Bootstrap that collapses into a hamburger menu on smaller screens using the collapse component.

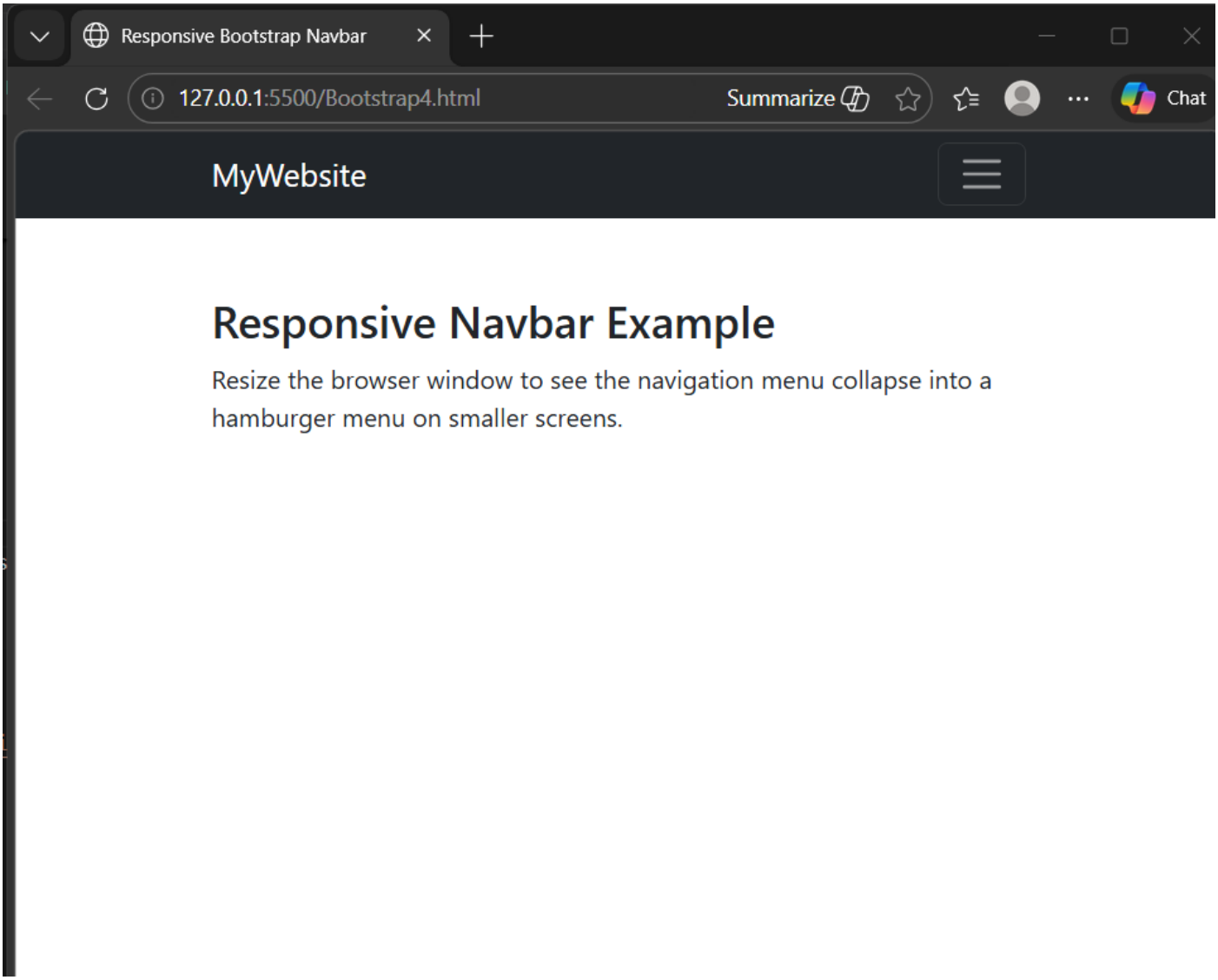
HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Responsive Bootstrap Navbar</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<nav class="navbar navbar-expand-lg navbar-dark bg-dark">
<div class="container">

<a class="navbar-brand" href="#">MyWebsite</a>

<button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-target="#navbarMenu">

<span class="navbar-toggler-icon"></span>

</button>

<div class="collapse navbar-collapse" id="navbarMenu">

<ul class="navbar-nav ms-auto">
```

```
<li class="nav-item">
<a class="nav-link active" href="#">Home</a>
</li>

<li class="nav-item">
<a class="nav-link" href="#">About</a>
</li>

<li class="nav-item">
<a class="nav-link" href="#">Services</a>
</li>

<li class="nav-item">
<a class="nav-link" href="#">Gallery</a>
</li>

<li class="nav-item">
<a class="nav-link" href="#">Contact</a>
</li>

</ul>
</div>
</div>
</nav>
<div class="container mt-5">

<h2>Responsive Navbar Example</h2>

<p>
Resize the browser window to see the navigation menu collapse into a hamburger menu on smaller screens.
</p>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>
```

Output:


Responsive Bootstrap Navbar
127.0.0.1:5500/Bootstrap4.html
Chat
MyWebsite
Home
About
Services
Gallery
Contact
Responsive Navbar Example
Resize the browser window to see the navigation menu collapse into a hamburger menu on smaller screens.


Responsive Bootstrap Navbar
127.0.0.1:5500/Bootstrap4.html
Summarize
Chat
MyWebsite
Responsive Navbar Example
Resize the browser window to see the navigation menu collapse into a hamburger menu on smaller screens.

5. Write a program to create a button that opens a modal window containing a login form. Also, add a tooltip to one of the input fields to give a hint.

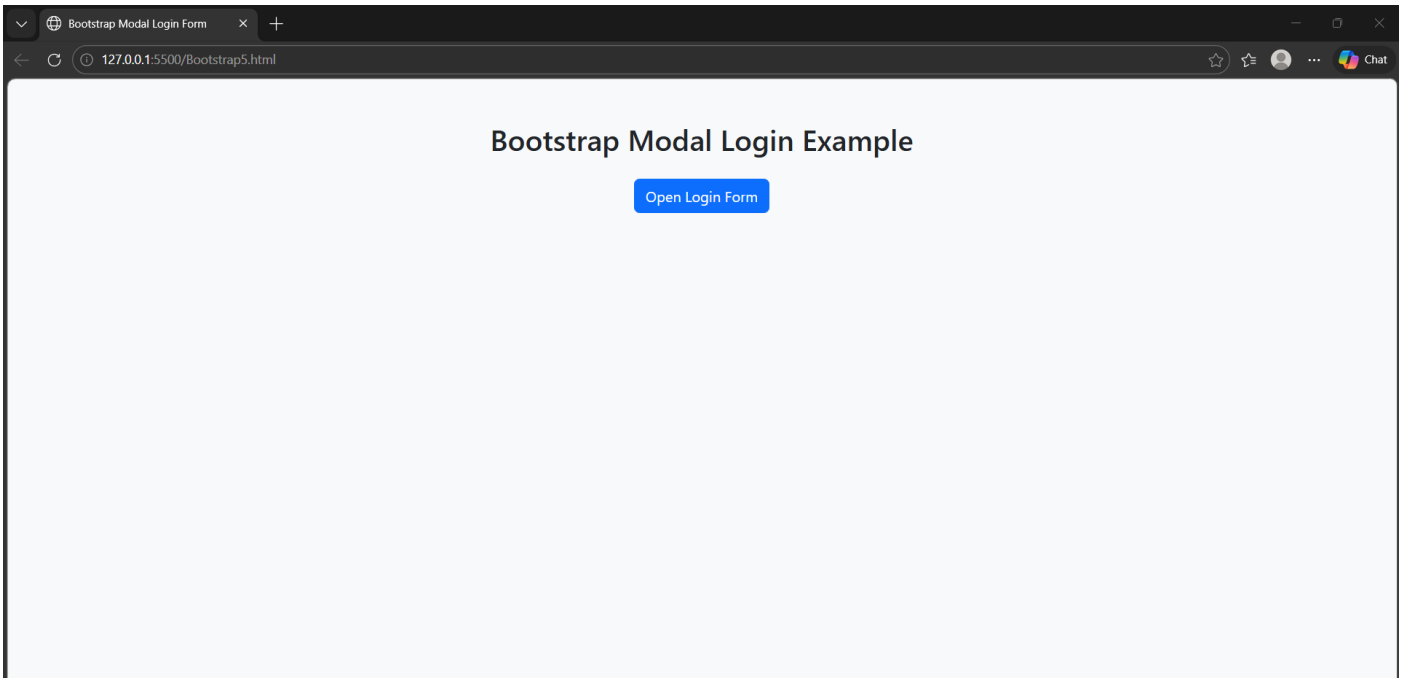
HTML Code:

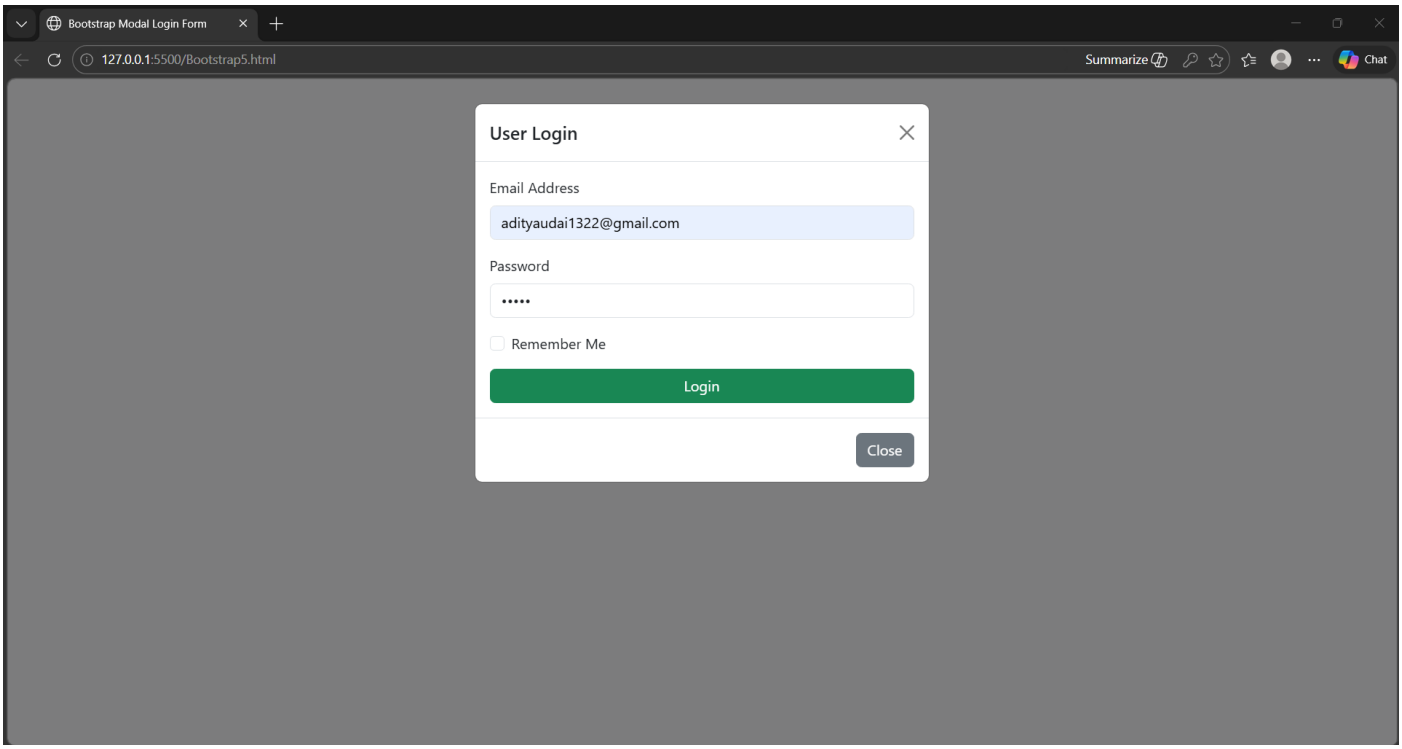
```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Bootstrap Modal Login Form</title>
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
<body class="bg-light">

<div class="container text-center mt-5">

<h2 class="mb-4">Bootstrap Modal Login Example</h2>
<button type="button" class="btn btn-primary" data-bs-toggle="modal" data-bs-target="#loginModal">
Open Login Form
</button>

</div>
<div class="modal fade" id="loginModal" tabindex="-1">

<div class="modal-dialog">
<div class="modal-content">
<div class="modal-header">

<h5 class="modal-title">User Login</h5>

<button type="button" class="btn-close" data-bs-dismiss="modal"></button>

</div>
<div class="modal-body">

<form>
<div class="mb-3">

<label class="form-label">Email Address</label>

<input type="email"
class="form-control"
placeholder="Enter your email"
```

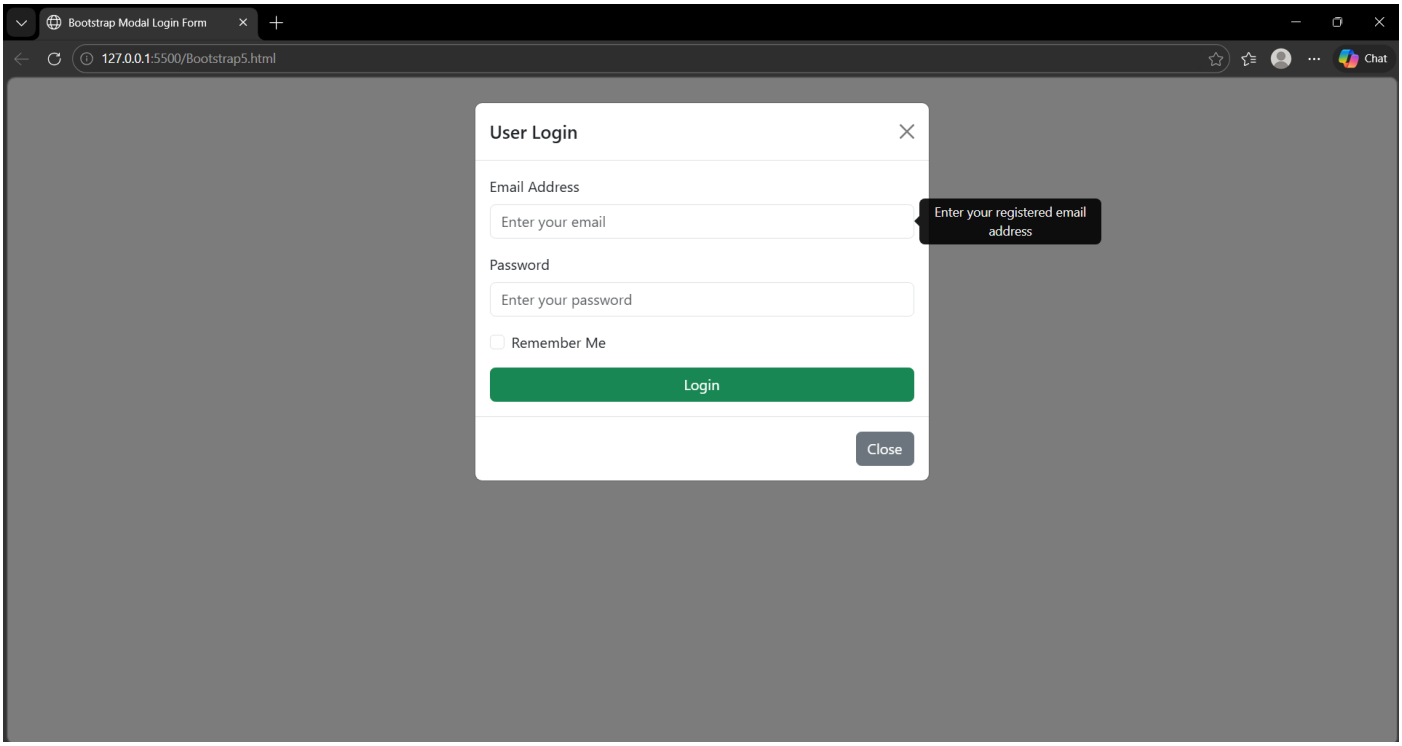
```
data-bs-toggle="tooltip"
data-bs-placement="right"
title="Enter your registered email address">

</div>
<div class="mb-3">

<label class="form-label">Password</label>

<input type="password"
class="form-control"
placeholder="Enter your password">

</div>
<div class="form-check mb-3">

<input type="checkbox" class="form-check-input">

<label class="form-check-label">Remember Me</label>

</div>

<button type="submit" class="btn btn-success w-100">
Login
</button>
</form>
</div>
<div class="modal-footer">

<button class="btn btn-secondary" data-bs-dismiss="modal">
Close
</button>

</div>

</div>
</div>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>

<script>

var tooltipTriggerList = document.querySelectorAll('[data-bs-toggle="tooltip"]')

var tooltipList = [...tooltipTriggerList].map(function (tooltipTriggerEl) {
```

```
return new bootstrap.Tooltip(tooltipTriggerEl)
})

</script>

</body>
</html>
```

Output:







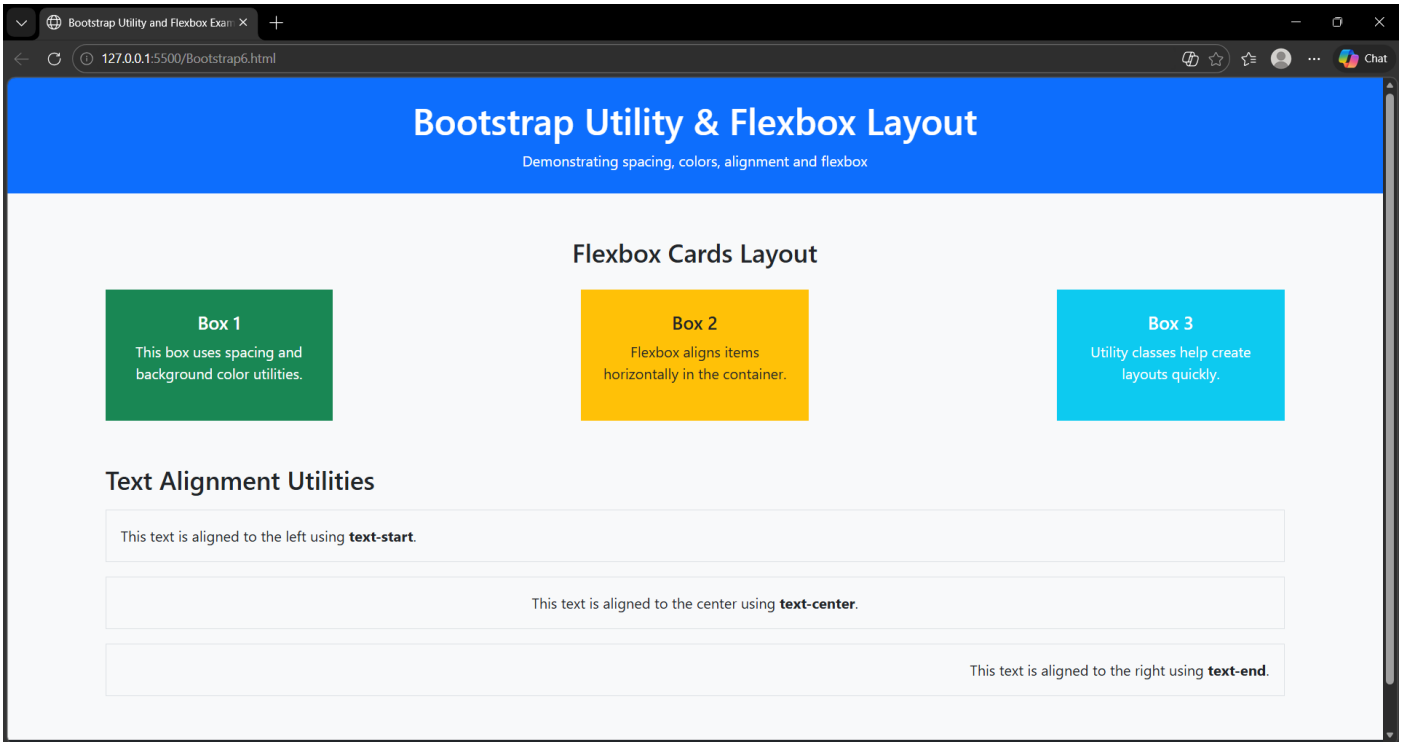
6. Write a program to demonstrate the use of Bootstrap's utility classes (for spacing, colors, and text alignment) along with flexbox classes (d-flex, justify-content, align-items) to build a flexible layout.

HTML Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Bootstrap Utility and Flexbox Example</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

</head>

<body class="bg-light">

<div class="bg-primary text-white text-center p-4">
<h1>Bootstrap Utility & Flexbox Layout</h1>
<p class="mb-0">Demonstrating spacing, colors, alignment and flexbox</p>
</div>

<div class="container my-5">

<h3 class="text-center mb-4">Flexbox Cards Layout</h3>
<div class="d-flex justify-content-between align-items-center flex-wrap gap-3">
```

```
<div class="bg-success text-white p-4 text-center" style="width:250px;">
<h5>Box 1</h5>
<p>This box uses spacing and background color utilities.</p>
</div>

<div class="bg-warning text-dark p-4 text-center" style="width:250px;">
<h5>Box 2</h5>
<p>Flexbox aligns items horizontally in the container.</p>
</div>

<div class="bg-info text-white p-4 text-center" style="width:250px;">
<h5>Box 3</h5>
<p>Utility classes help create layouts quickly.</p>
</div>
</div>
</div>
<div class="container">

<h3 class="mb-3">Text Alignment Utilities</h3>

<p class="text-start bg-light p-3 border">
This text is aligned to the left using <b>text-start</b>.
</p>

<p class="text-center bg-light p-3 border">
This text is aligned to the center using <b>text-center</b>.
</p>

<p class="text-end bg-light p-3 border">
This text is aligned to the right using <b>text-end</b>.
</p>

</div>
<div class="bg-dark text-white text-center p-3 mt-5">
<p class="mb-0">© 2026 Bootstrap Flexbox Demo</p>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```

Output:



7. Write a program to add an image carousel to a webpage using Bootstrap and use badges to highlight a featured or new product.

HTML Code

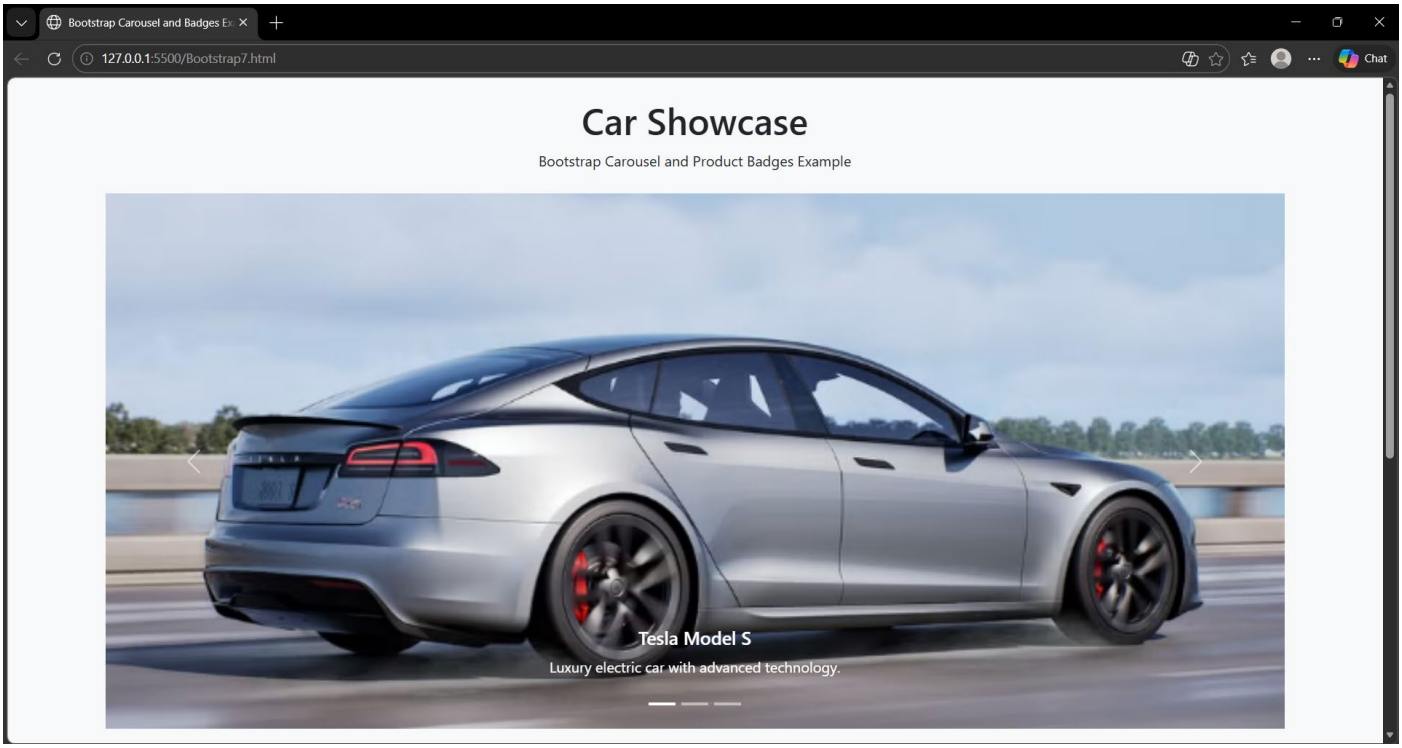
```
<!DOCTYPE html>
<html lang="en">

<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>Bootstrap Carousel and Badges Example</title>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

</head>

<body class="bg-light">

<div class="container text-center my-4">
<h1>Car Showcase</h1>
<p>Bootstrap Carousel and Product Badges Example</p>
</div>

<div class="container mb-5">

<div id="carCarousel" class="carousel slide" data-bs-ride="carousel">
<div class="carousel-indicators">
<button type="button" data-bs-target="#carCarousel" data-bs-slide-to="0" class="active"></button>
```

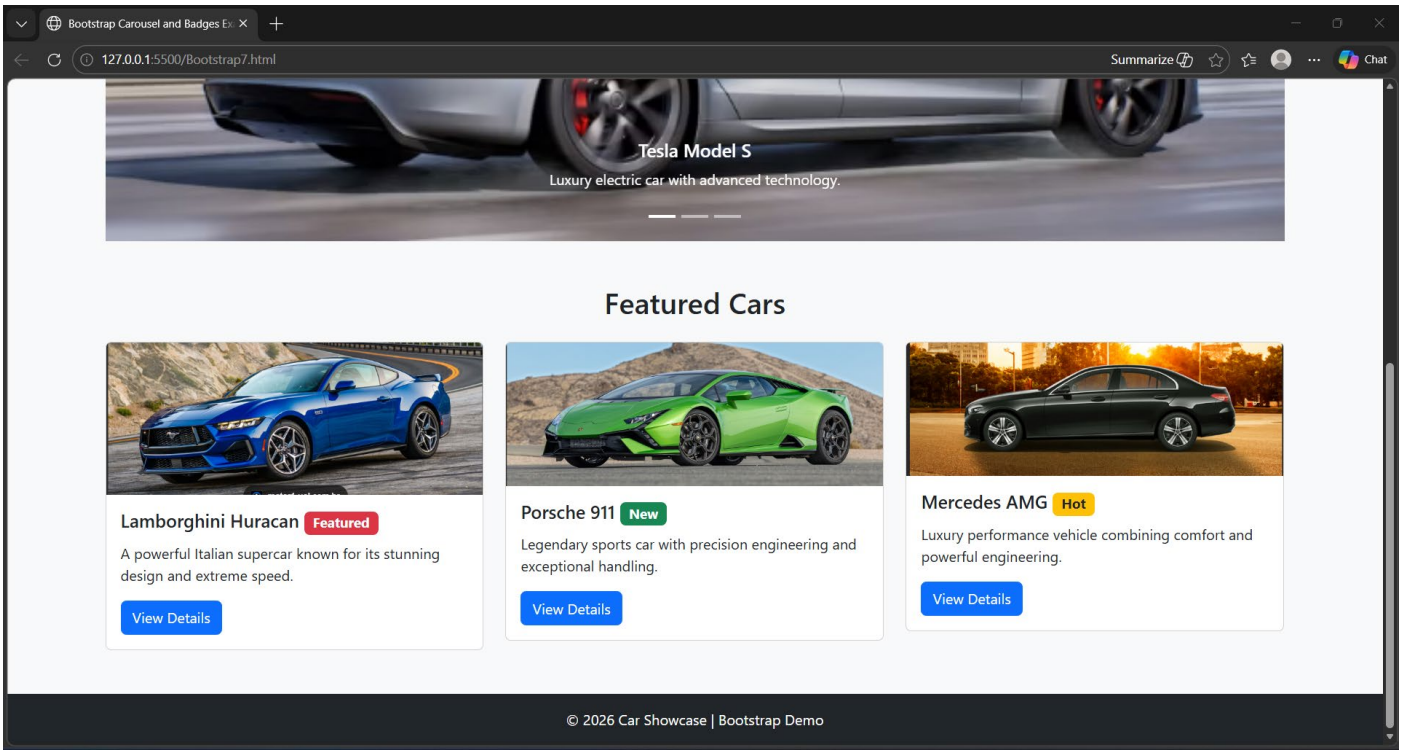
```
<button type="button" data-bs-target="#carCarousel" data-bs-slide-to="1"></button>
<button type="button" data-bs-target="#carCarousel" data-bs-slide-to="2"></button>
</div>
<div class="carousel-inner">

<div class="carousel-item active">
<img src="1.jpg" class="d-block w-100" alt="Tesla Model S">
<div class="carousel-caption d-none d-md-block">
<h5>Tesla Model S</h5>
<p>Luxury electric car with advanced technology.</p>
</div>
</div>

<div class="carousel-item">
<img src="2.jpg" class="d-block w-100" alt="BMW M4">
<div class="carousel-caption d-none d-md-block">
<h5>BMW M4</h5>
<p>High performance sports coupe.</p>
</div>
</div>

<div class="carousel-item">
<img src="3.jpg" class="d-block w-100" alt="Audi R8">
<div class="carousel-caption d-none d-md-block">
<h5>Audi R8</h5>
<p>Supercar with incredible power and design.</p>
</div>
</div>

</div>
<button class="carousel-control-prev" type="button" data-bs-target="#carCarousel" data-bs-slide="prev">
<span class="carousel-control-prev-icon"></span>
</button>

<button class="carousel-control-next" type="button" data-bs-target="#carCarousel" data-bs-slide="next">
<span class="carousel-control-next-icon"></span>
</button>

</div>

</div>
<div class="container">

<h2 class="text-center mb-4">Featured Cars</h2>
```

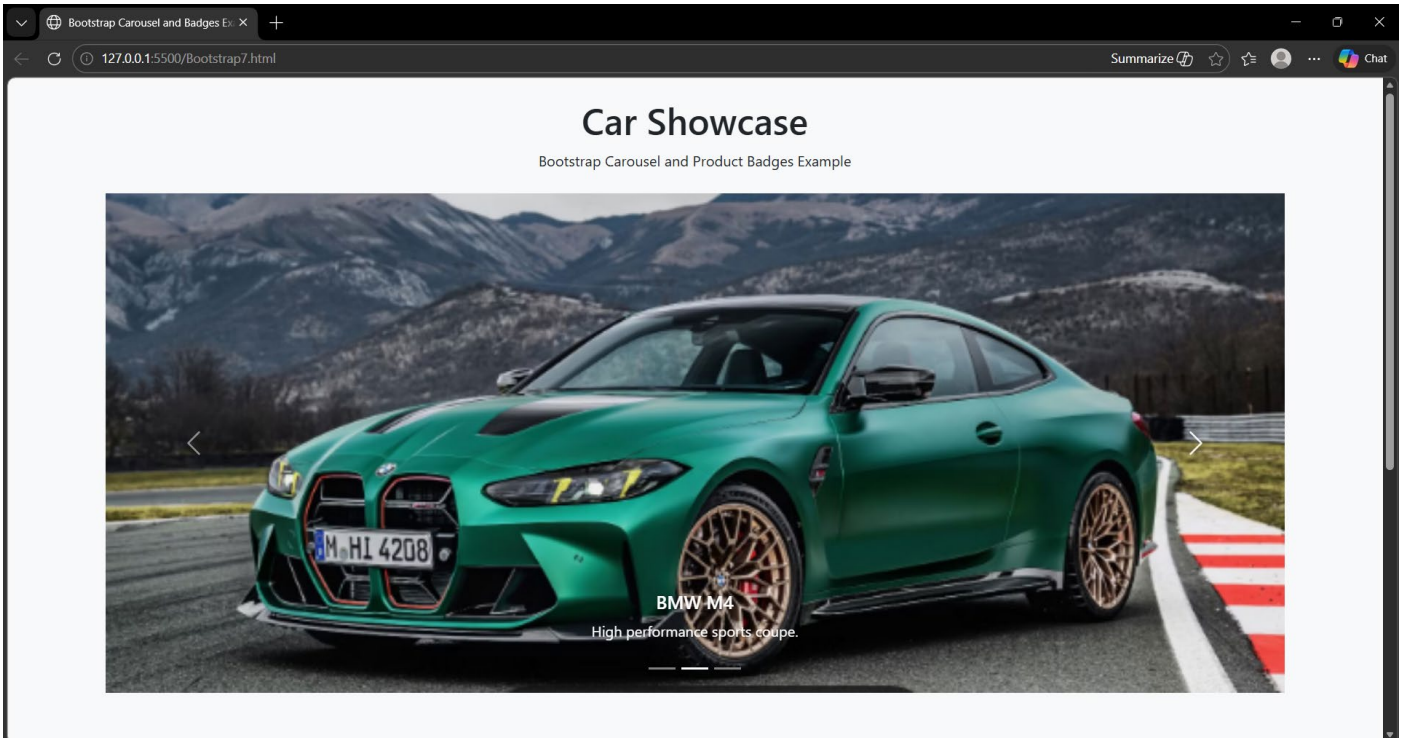
```
<div class="row g-4">

<div class="col-md-4">
<div class="card">

<img src="4.jpg" class="card-img-top">

<div class="card-body">

<h5 class="card-title">
Lamborghini Huracan
<span class="badge bg-danger">Featured</span>
</h5>

<p class="card-text">
A powerful Italian supercar known for its stunning design and extreme speed.
</p>

<a href="#" class="btn btn-primary">View Details</a>

</div>
</div>
</div>
<div class="col-md-4">
<div class="card">

<img src="5.jpg" class="card-img-top">

<div class="card-body">

<h5 class="card-title">
Porsche 911
<span class="badge bg-success">New</span>
</h5>

<p class="card-text">
Legendary sports car with precision engineering and exceptional handling.
</p>

<a href="#" class="btn btn-primary">View Details</a>

</div>
</div>
</div>
<div class="col-md-4">
```

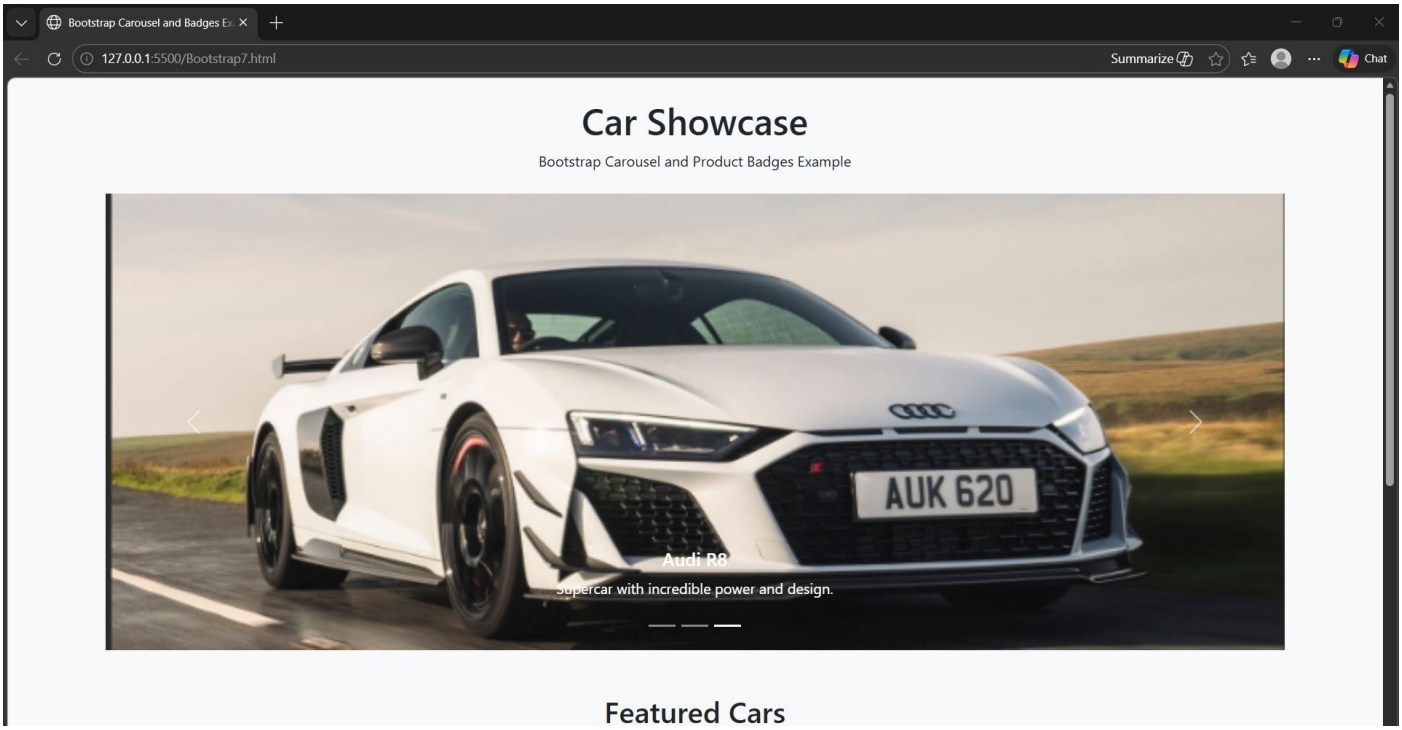
```
<div class="card">

<img src="6.jpg" class="card-img-top">

<div class="card-body">

<h5 class="card-title">
Mercedes AMG
<span class="badge bg-warning text-dark">Hot</span>
</h5>

<p class="card-text">
Luxury performance vehicle combining comfort and powerful engineering.
</p>

<a href="#" class="btn btn-primary">View Details</a>

</div>
</div>
</div>
</div>
</div>
<div class="bg-dark text-white text-center p-3 mt-5">
<p class="mb-0">© 2026 Car Showcase | Bootstrap Demo</p>
</div>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"></script>

</body>
</html>
```

Output:


Bootstrap Carousel and Badges Ex
127.0.0.1:5500/Bootstrap7.html
Chat
Car Showcase
Bootstrap Carousel and Product Badges Example
Tesla Model S
Luxury electric car with advanced technology.


Bootstrap Carousel and Badges Ex
127.0.0.1:5500/Bootstrap7.html
Summarize
Chat
Tesla Model S
Luxury electric car with advanced technology.
Featured Cars
Lamborghini Huracan Featured
A powerful Italian supercar known for its stunning design and extreme speed.
View Details
Porsche 911 New
Legendary sports car with precision engineering and exceptional handling.
View Details
Mercedes AMG Hot
Luxury performance vehicle combining comfort and powerful engineering.
View Details
© 2026 Car Showcase | Bootstrap Demo


Bootstrap Carousel and Badges Ex
127.0.0.1:5500/Bootstrap7.html
Summarize
Chat
Car Showcase
Bootstrap Carousel and Product Badges Example
M HI 4208
BMW M4
High performance sports coupe.


Bootstrap Carousel and Badges Ex
127.0.0.1:5500/Bootstrap7.html
Summarize
Chat
Car Showcase
Bootstrap Carousel and Product Badges Example
AUK 620
Audi R8
Supercar with incredible power and design.
Featured Cars